

# Lecture 3

## The Singular Value Decomposition

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Introduction to Numerical Methods

Per-Olof Persson (persson@mit.edu)

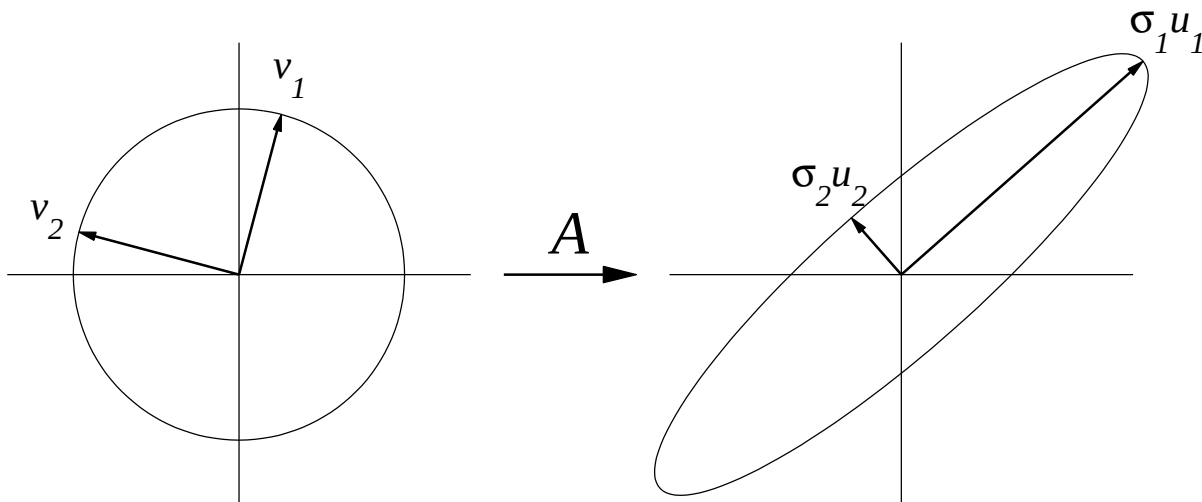
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## The SVD - The Main Idea

- Motivation:

**The image of the unit sphere under any  $m \times n$  matrix is a hyperellipse**



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## The SVD - Brief Description

- Suppose (for the moment) that  $A$  is  $m \times n$  with  $m \geq n$  and full rank  $n$
- Choose orthonormal bases

$v_1, \dots, v_n$  for the row space

$u_1, \dots, u_n$  for the column space

such that  $Av_i$  is in the direction of  $u_i$ :

$$Av_i = \sigma_i u_i$$

- In MATLAB: `eigshow`
- The *singular values*  $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n > 0$

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## The SVD - Brief Description

- In Matrix form,  $Av_i = \sigma_i u_i$  becomes

$$AV = \hat{U}\hat{\Sigma}, \text{ that is, } \boxed{A = \hat{U}\hat{\Sigma}V^*}$$

where  $\hat{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_n)$

- This is the *reduced singular value decomposition*
- Add orthogonal extension to  $\hat{U}$  and add rows to  $\hat{\Sigma}$  to obtain the *full singular value decomposition*

$$\boxed{A = U\Sigma V^*}$$

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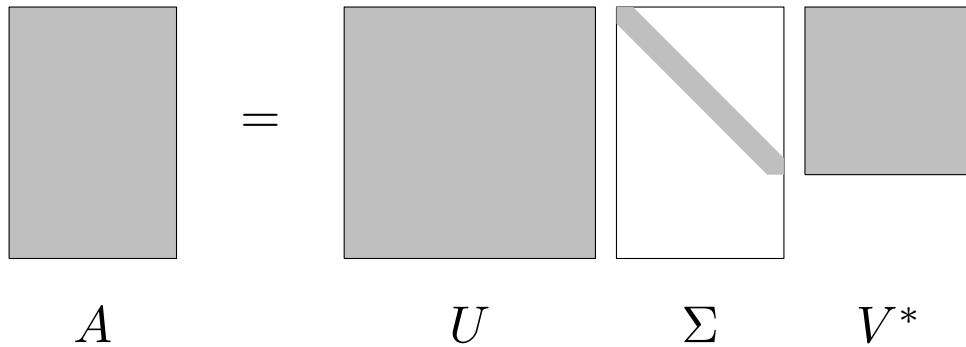
## The Full Singular Value Decomposition

- Let  $A$  be an  $m \times n$  matrix. The *singular value decomposition* of  $A$  is the factorization  $A = U\Sigma V^*$  where

$U$  is  $m \times m$  unitary (the left singular vectors of  $A$ )

$V$  is  $n \times n$  unitary (the right singular vectors of  $A$ )

$\Sigma$  is  $m \times n$  diagonal (the singular values of  $A$ )



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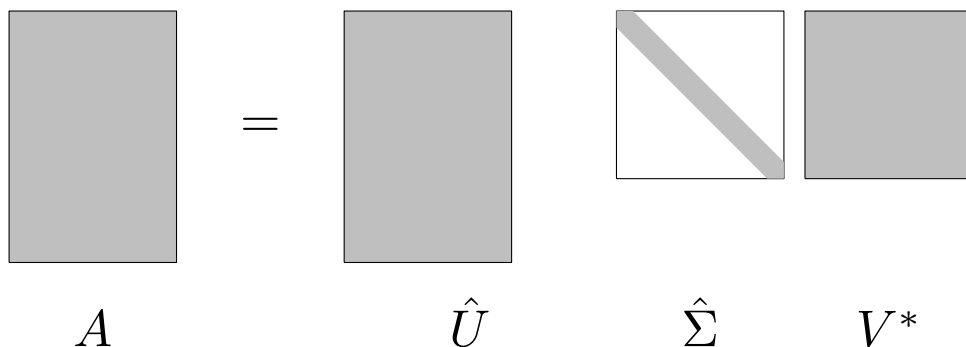
## The Reduced Singular Value Decomposition

- A more compact representation is the *Reduced SVD*, for  $m \geq n$ :

$$A = \hat{U}\hat{\Sigma}V^*$$

where

$\hat{U}$  is  $m \times n$ ,  $V$  is  $n \times n$ , and  $\Sigma$  is  $n \times n$



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## The SVD and The Eigenvalue Decomposition

- The *eigenvalue decomposition*  $A = X\Lambda X^{-1}$ 
  - uses the same basis  $X$  for row and column space, but the SVD uses two different bases  $V, U$
  - generally does not use an orthonormal basis, but the SVD does
  - is only defined for square matrices, but the SVD exists for all matrices
- For *symmetric positive definite* matrices  $A$ , the eigenvalue decomposition and the SVD are equal

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## Matrix Properties

1. The rank of  $A$  is  $r$ , the number of nonzero singular values
2.  $\text{range}(A) = \langle u_1, \dots, u_r \rangle$  and  $\text{null}(A) = \langle v_{r+1}, \dots, v_n \rangle$
3.  $\|A\|_2 = \sigma_1$  and  $\|A\|_F = \sqrt{\sigma_1^2 + \sigma_2^2 + \dots + \sigma_r^2}$
4. Nonzero eigenvalues of  $A^*A$  are nonzero  $\sigma_i^2$ , eigenvectors are  $v_i$   
Nonzero eigenvalues of  $AA^*$  are nonzero  $\sigma_i^2$ , eigenvectors are  $u_i$
5. If  $A = A^*$ ,  $\sigma_i = |\lambda_i|$  where  $\lambda_i$  are eigenvalues of  $A$
6. For square  $A$ ,  $|\det(A)| = \prod_{i=1}^m \sigma_i$

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## Existence and Uniqueness

- Every matrix has a singular value decomposition
- The singular values  $\sigma_j$  are uniquely determined
- If  $A$  square and  $\sigma_j$  distinct, left/right singular vectors  $u_j, v_j$  are uniquely determined up to complex signs
- *Proof.* Textbook / Black board

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## Low-Rank Approximations

- The SVD can be written as a sum of rank-one matrices

$$A = \sum_{j=1}^r \sigma_j u_j v_j^*$$

- The best rank  $\nu$  approximation of  $A$  in the 2-norm is

$$A_\nu = \sum_{j=1}^{\nu} \sigma_j u_j v_j^*$$

with  $\|A - A_\nu\|_2 = \sigma_{\nu+1}$

- Also true in the Frobenius norm, with  $\|A - A_\nu\|_2 = \sqrt{\sigma_{\nu+1}^2 + \cdots + \sigma_r^2}$

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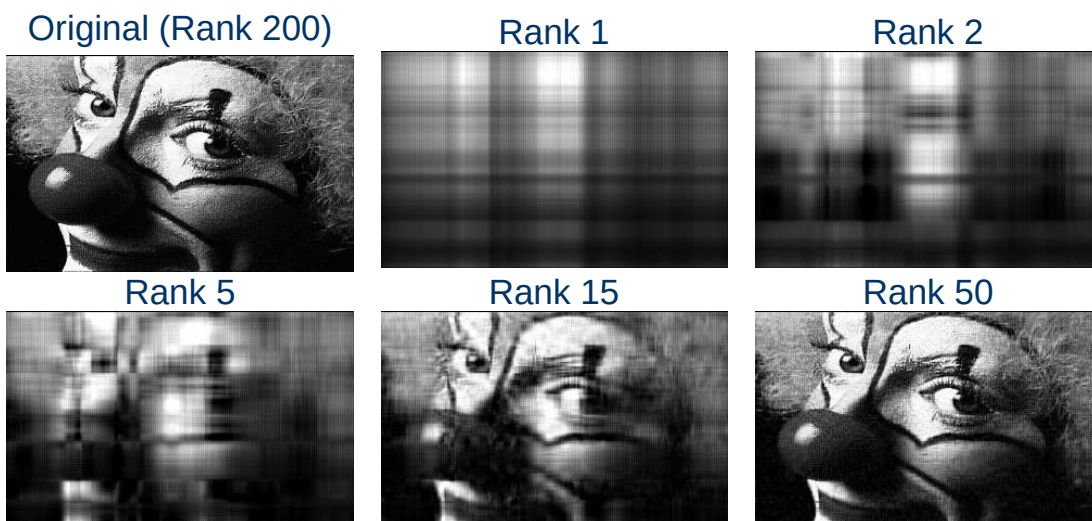
## Applications of the SVD

- Calculation of matrix properties:
  - Rank of matrix (counting  $\sigma_j$ 's  $>$  tolerance)
  - Bases for range and nullspace (in  $U$  and  $V$ )
  - Induced matrix norm  $\| \cdot \|_2$  ( $= \sigma_1$ )
- Low-rank approximations (optimal in  $\| \cdot \|_2$  and  $\| \cdot \|_F$ )
- Least squares fitting (more later, another option is  $QR$ )
- Signal and image processing
  - Compression (see next slide)
  - Noise removal (noise tends to have low  $\sigma_j$ )

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## Application: Image Compression

- View  $m \times n$  image as a (real) matrix  $A$ , find best rank  $\nu$  approx. by SVD
- Storage  $\nu(m + n)$  instead of  $mn$



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